Dynamic Programming: From Local Optimality to Global Optimality

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August 20, 2025

Introduction

DP notation:

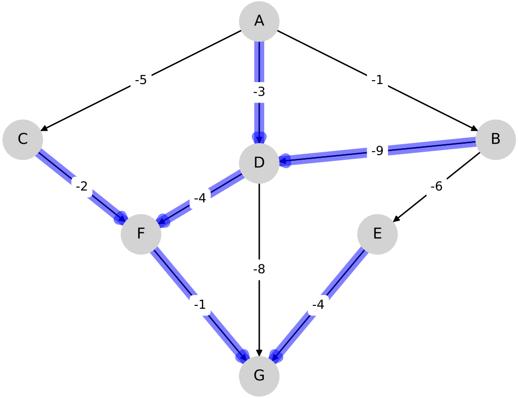
- X is the **state space**
- A is the **action space**
- A **policy** is a map $\sigma \colon \mathsf{X} \to \mathsf{A}$
- $\sigma(x)$ = action in state x

A **feasible policy** is a policy satisfying some feasibility constraint

We let Σ denote the **set of all feasible policies**

Eg. Traversing a graph

- X = all nodes and A = all edges
- σ maps each node to an edge



Let $v_\sigma(x)$ = **lifetime value** when

1. following policy σ forever
2. starting from initial condition $x \in \mathsf{X}$

Eg. In the optimal savings problem

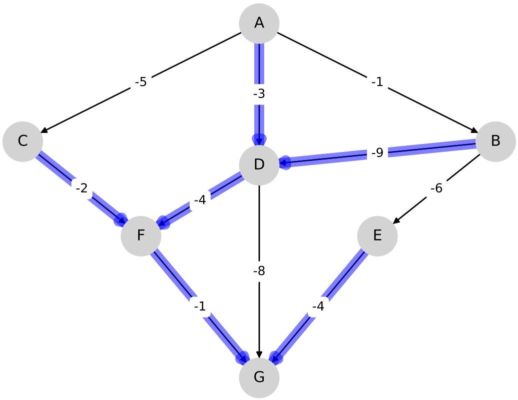
$$v_\sigma(x) = \mathbb{E} \sum_{t \geqslant 0} \beta^t u(C_t)$$

where

- consumption is determined by σ and
- initial wealth is x

Eg. $v_\sigma(x)$ = reward from traversing graph starting at node x

- $v_\sigma(A) = -8$



The **value function** is

$$v^*(x) \coloneqq \sup_{\sigma \in \Sigma} v_\sigma(x)$$

Def. Policy σ is called **optimal** when

$$v_\sigma = v^*$$

Equivalent:

$$v_s \leqslant v_\sigma \text{ for all } s \in \Sigma$$

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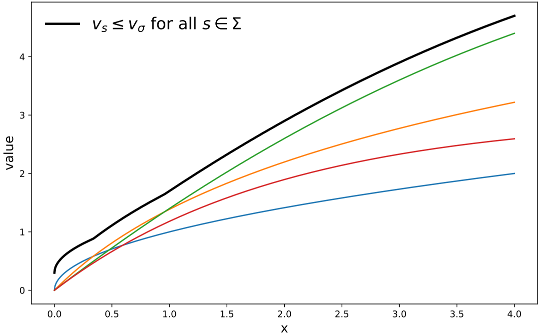
In other words, we seek a **greatest element** in a **partially ordered set**



Our research question

Recall: σ is optimal when

$$v_\sigma(x) = v^*(x) \text{ for } \textbf{every } x \in \mathsf{X}$$

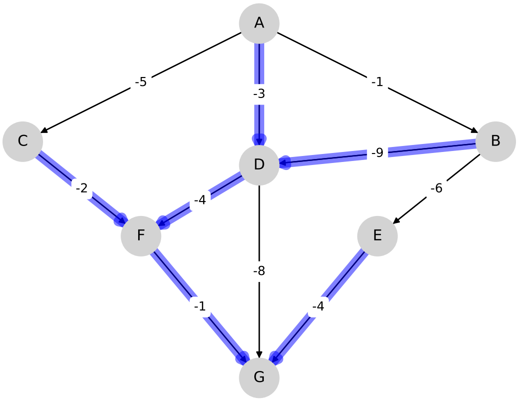
Our question: Under what conditions do we have

$$v_\sigma(x) = v^*(x) \text{ for } \textbf{some } x \in \mathsf{X} \quad \Longrightarrow \quad v_\sigma(x) = v^*(x) \text{ for } \textbf{every } x \in \mathsf{X}$$

When does "local optimality" imply "global optimality"?

Not always true, of course

- $v_{\sigma}(A) = v^*(A)$ but not $v_{\sigma}(B) = v^*(B)$



When might it be true?

- Say we freeze x and solve $\max_\sigma v_\sigma(x)$
- The value $v_\sigma(x)$ depends on present and future rewards
- Hence making $v_\sigma(x)$ large requires that σ does well in the future
- Therefore, **σ should perform well in states we are likely to visit**
- If we visit all of X under σ, then σ should perform well everywhere

Possible conclusion:

Sufficient mixing under $\sigma \implies$

local optimality implies global optimality

Motivation

1. Better understanding of theoretical properties
2. Assist modern solution methods for high-dimensional DP problems

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Example: Policy gradient methods

1. Fix an **initial distribution** ρ and set

$$F(\boldsymbol{\sigma}) := \int v_{\boldsymbol{\sigma}}(x)\rho(\mathrm{d}x)$$

2. Maximize F over Σ using **gradient ascent**:

$$\boldsymbol{\sigma}_{n+1} = \boldsymbol{\sigma}_n + \lambda_n \nabla F(\boldsymbol{\sigma}_n)$$

- ANNs, autodiff, backprop, 1000s of GPUs, etc.

Key point: Fixing an initial distribution $\implies$ **real-valued objective**

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$$F(\sigma) = \int v_\sigma(x)\rho(\mathrm{d}x)$$

But does maximizing $F(\sigma) = \int v_{\sigma}(x)\rho(\mathrm{d}x)$ lead to optimality?

Special case: does maximizing $F(\sigma) = v_{\sigma}(\bar{x})$ lead to optimality?

In other words, **does local optimality imply global optimality?**

- Are there settings where this holds true?

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Setup

Consider a DP problem with **Bellman equation**

$$v(x) = \max_{a \in \Gamma(x)} \left\{ r(x, a) + \beta \int v(x') P(x, a, \mathrm{d}x') \right\}$$

Here

- $\beta \in (0, 1)$
- Some mild continuity-type restrictions (for existence)

For given σ we let

$$P_\sigma(x, \mathrm{d}x') = P(x, \sigma(x), \mathrm{d}x') = \textbf{Markov kernel under } \sigma$$

Standard optimality results hold

Proposition 1

Under mild assumptions,

1. At least one optimal policy exists
2. v^* is the unique solution to the Bellman equation
3. A policy σ is optimal if and only if

$$\sigma(x) \in \operatorname*{arg\,max}_{a \in \Gamma(x)} \left\{ r(x, a) + \beta \int v^*(x') P(x, a, \mathrm{d}x') \right\} \quad \text{for all } x \in \mathsf{X}$$

4. Value function iteration converges to v^*

Our results

Fix $\sigma \in \Sigma$ and consider the following statements:

(E1) There exists an $x \in \mathsf{X}$ such that $v_\sigma(x) = v^*(x)$

(E2) There exists a distribution ρ on X such that

$$\int v_\sigma(x)\rho(\mathrm{d}x) = \int v^*(x)\rho(\mathrm{d}x)$$

(E3) σ is an optimal policy

We seek conditions under which (E1)–(E3) are equivalent

Irreducibility (Banach lattice setting)

Def. Let

- E be a Banach lattice
- E' be the dual space
- E_+ and E'_+ be the respective positive cones

A positive linear operator K on E is called **irreducible** if

- for all nonzero $f \in E_+$
- and all nonzero $\mu \in E'_+$

there exists an $n \in \mathbb{N}$ with $\langle \mu, K^n f \rangle > 0$

Theorem 1: Global Optimality and Irreducibility

If P_σ is irreducible, then (E1)–(E3) are equivalent

Implications:

- Maximizing $F(\sigma) = \int v_\sigma(x)\rho(\mathrm{d}x)$ **always** produces an optimal σ
- Maximizing $v_\sigma(x)$ at fixed x **always** produces an optimal policy

Intuition:

- Irreducibility under σ supplies the mixing

Theorem 1: Global Optimality and Irreducibility

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Example: optimal saving

Wealth obeys

$$W_{t+1} = R_{t+1}(W_t - C_t) + Y_{t+1}$$

where

- (Y_t) and (R_t) are IID with distributions φ and ψ
- $\mathsf{X} = \mathsf{A} = \mathbb{R}_+$
- a policy is a map σ with $0 \leqslant \sigma(w) \leqslant w$ for all $w \in \mathbb{R}_+$

Assume u is bounded, increasing, continuous, and strictly concave

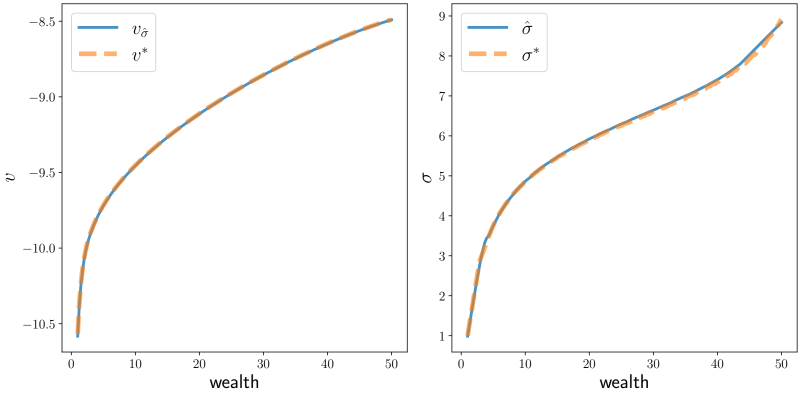

Figure 1: Irreducible case: $v_{\hat{\sigma}}$ and $\hat{\sigma}$ with $\bar{w} = 1$ vs OPI solutions

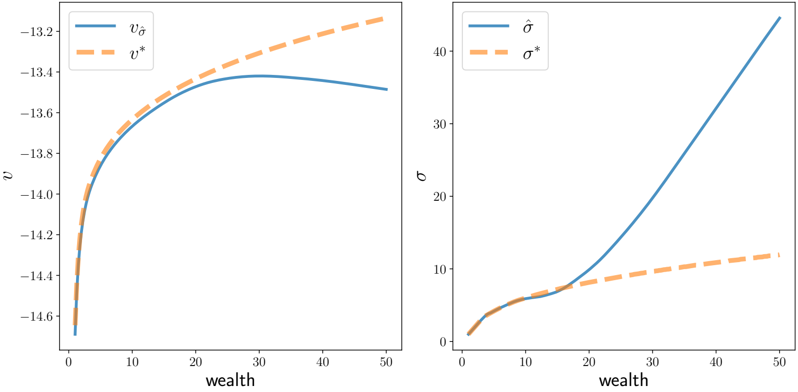

Figure 2: Reducible case: $\hat{v}_\sigma$ and $\hat{\sigma}$ with $\bar{w} = 1$ against the OPI solutions.

Questions

What about

- "Partial" irreducibility?
- State-dependent discounting?
- Recursive preference models?
- Continuous time MDPs?
- Dynamic games?